

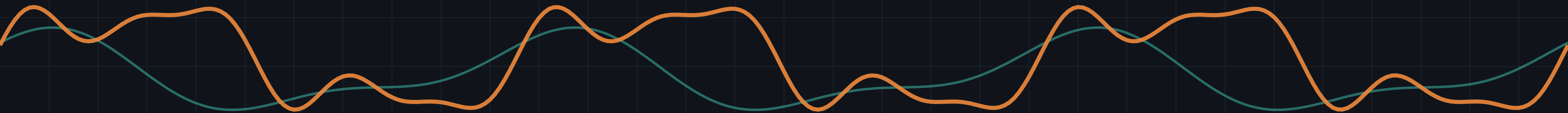
CSE 220 · SIGNALS AND LINEAR SYSTEMS · PROJECT

WaveKitchen

A cooking game where every kitchen action is a signal-processing operation

Generate · Wash · Mix · Season · Marinate · Cook · Oven · Deliver

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MOTIVATION

Equations you can see, hear and score

Convolution

is a formula on the board, not a sound you hear.

Aliasing

is a warning in the textbook, not a mistake you make.

Poles and zeros

are dots on a plane, not a filter you tune.

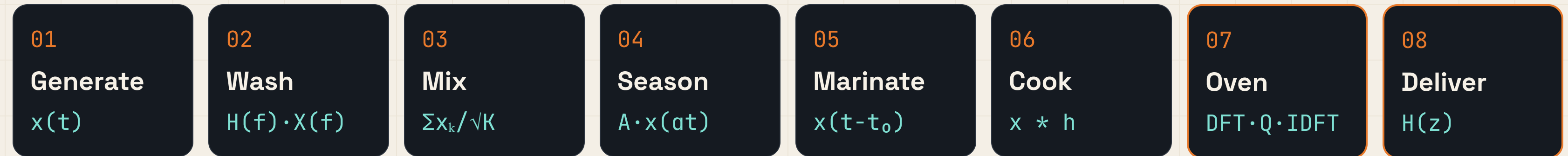
THE GAME AS A SYSTEM

Make your dish match a hidden reference dish.

```
ingredient →  $x(t)$ 
station     → a system  $T\{\cdot\}$ 
dish        →  $y(t) = T_8\{ \dots T_2\{ T_1\{x(t)\} \} \}$ 
target      →  $y_{\text{ref}}(t)$ , the same chain
              on clean ingredients
score       →  $\text{similarity}(y, y_{\text{ref}})$ , 0-100
```

APPROACH

One dish, one signal chain



THE SERVED DISH, END TO END

```
m = (1/√K) · Σ_k H_wash{ x_k }      (wash, then mix)
c = A · m(α(t - t_o)) * h + b(t)      (season, marinate, cook)
y = H_cart{ IDFT{ Q · DFT{c} } + v(t) } (oven, then cart)
```

b = burnt overtone, v = road vibration. The reference y_{ref} is this same chain on clean ingredients with the recipe's ideal dials. 5 recipes · 3 tiers · 4 difficulties (7:00 → 2:00).

Generate, wash, mix

01 · GENERATE

Ingredients are signals

$$x[n] = x(n/400), \quad n = 0 \dots 400$$

$$\text{sq} = (4A/\pi) \cdot \sum \sin(2\pi kft)/k$$

$$\text{tri} = (8A/\pi^2) \cdot \sum \pm \sin(2\pi kft)/k^2$$

$$k = 1, 3, 5, \dots \quad (\text{odd harmonics})$$

$$x_d(t) = x(t) + n(t)$$

Parametric curves use $y(t)$; the chicken is a real 44.1 kHz recording.

02 · WASH

Washing is low-pass filtering

$$Y(f) = H(f) \cdot X(f)$$

$$H = 1, \quad \text{if } f \leq f_c$$

$$H = \frac{1}{2}[1 + \cos(\pi(f - f_c)/w)]$$

$$H = 0, \quad \text{if } f \geq f_c + w$$

$$w = \max(30, 0.15 \cdot f_c)$$

$$\text{prep} = 55\% \cdot \text{removed} + 45\% \cdot \text{kept}$$

Pass mark: cleanliness $\geq 85\%$.

03 · MIX

The bowl is superposition

$$m[n] = (1/\sqrt{K}) \cdot \sum_k x_k[n]$$

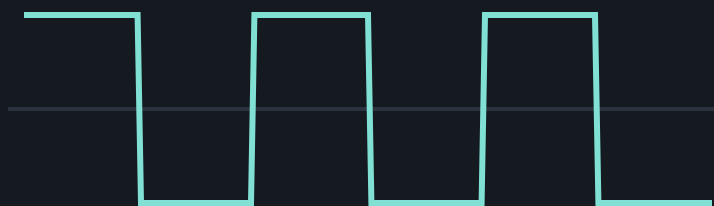
$$K = 1 \rightarrow m[n] = x_1[n]$$

$$\text{curve} = (\sum_k X_k(u), \sum_k Y_k(u))$$

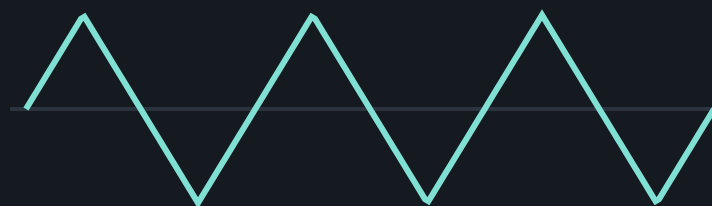
$$\text{mix\%} = |\text{bowl} \cap \text{recipe}| / |\text{recipe}|$$

$1/\sqrt{K}$ keeps K ingredients about as loud as one; curves add as 2-D paths.

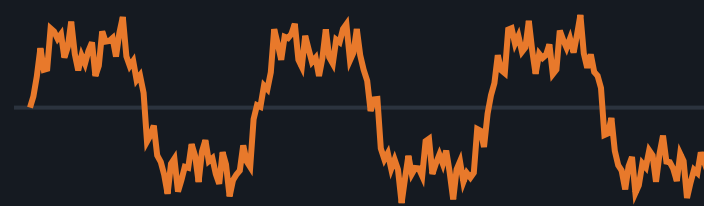
square: odd harmonics, $1/k$



triangle: odd harmonics, $1/k^2$



dirty: $x(t) + n(t)$



washed: $H \cdot X$, noise removed



Season, marinate, cook

04 · SEASON

Amplitude and time scaling

$$s(t) = A \cdot m(at)$$

$$S(f) = (A/|a|) \cdot M(f/a)$$

$a > 1$: compressed, higher f

$$\text{acc} = 100 - 70 \cdot (\Delta A/A^* + \Delta a/a^*)$$

No rescaling afterwards, so a level mistake stays visible.

05 · MARINATE

Time shift

$$r(t) = m(t - t_o)$$

$$R(f) = M(f) \cdot e^{-j2\pi f t_o}$$

$$d = \text{round}(t_o/3 \cdot N) \text{ samples}$$

$$\text{acc} = 100 - 150 \cdot \Delta t_o/t_o^*$$

The server delays by fractional samples with this linear phase ramp.

06 · COOK

Cooking is LTI convolution

$$y[n] = \sum_k x[(n-k-s) \bmod N] \cdot h[k]$$

$$k = 0..95, \quad \sum |h[k]| = 1$$

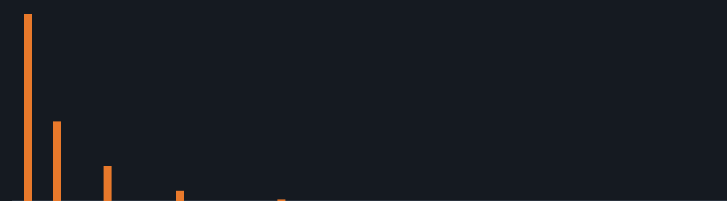
$$s = \text{round}(30\tau), \quad \tau = \text{depth}$$

$$\text{out} = (1 - w) \cdot x + w \cdot y$$

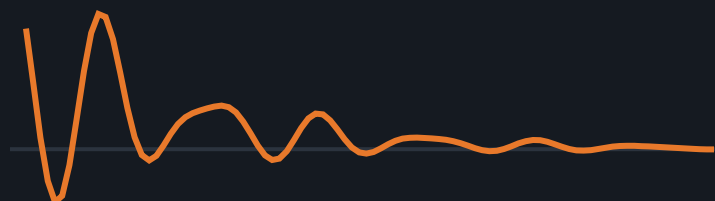
$$w = \min(1, 1.15\tau)$$

The impulse response $h[n]$ of each appliance is drawn below.

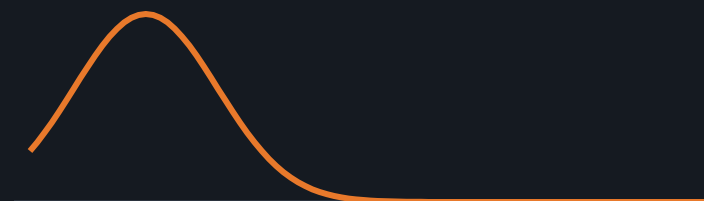
grill: an echo comb



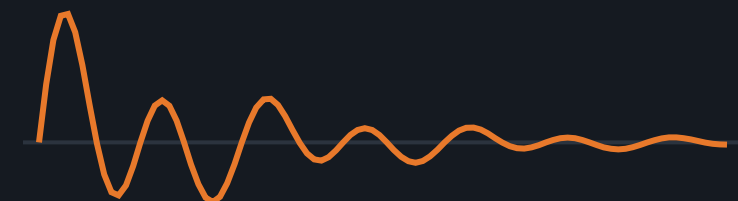
fry: a bright burst



bake: $e^{-((k-16)/14)^2}$



boil: a damped resonator



Precision Oven: sampling, DFT and the notch

07 · SAMPLE AND TRANSFORM

```

b[n] = a_d · max|x| · sin(2π · f_d · n / 400)
f_d = p · f_o    (between two harmonics)
f_s ≥ 2 · f_max  (Nyquist)
f_alias = |f - round(f/f_s) · f_s|
X[k] = Σ_m x[m] · e^(-j2πkm/N)
f_k = k · f_s / N

```

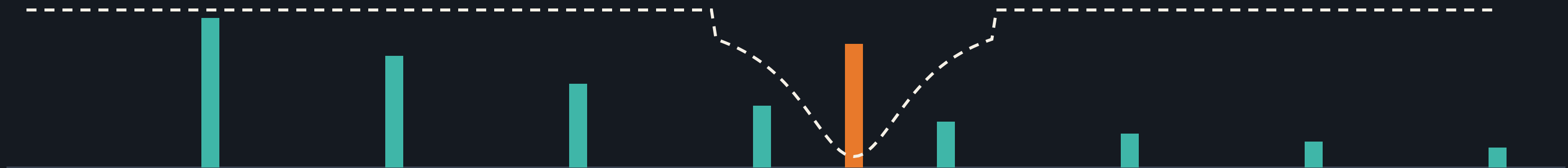
07 · FILTER AND REBUILD

```

X'[k] = X[k] · G_band · C(f_k) · Q(f_k)
Q(f) = 1 - 0.98 / (1 + ((f - f_n) / 1.5)²)
C(f) = e^(-((f - f_c) / 4)²), f > f_c
X'[N-k] = conj(X'[k])    (real output)
x'[m] = (1/N) · Σ_k X'[k] · e^(j2πkm/N)
x_out(t) = Σ_m x'[m] · D_N(t - m/N)

```

harmonics every 4 Hz · burnt overtone at 18 Hz · dashed: the notch gain $Q(f) · 0 \rightarrow 32$ game Hz



System Delivery: poles, zeros and stability

08 · THE CART AS A FILTER

$$v[n] = x[n] + a_r \cdot \max|x| \cdot \cos(\omega_r \cdot n)$$

$$\omega_r = 2\pi \cdot f_r / 8000 \quad (\text{burger: } 2600 \text{ Hz} \rightarrow 2.04 \text{ rad})$$

$$H(z) = g \cdot (z - e^{j\omega_o})(z - e^{-j\omega_o}) / (z - 0.85e^{j\omega_o})(z - 0.85e^{-j\omega_o})$$

$$y[n] = \sum_k b_k \cdot x[n-k] - \sum_k a_k \cdot y[n-k]$$

$$|H(e^{j\omega})| = g \cdot \prod_i |e^{j\omega} - z_i| / \prod_j |e^{j\omega} - p_j|$$

$$\text{stable} \Leftrightarrow \text{every } |p_j| < 1 \quad (\text{BIBO})$$

$$\text{low-pass: } H(z) = (1 - r) \cdot z / (z - r)$$

$$\text{sensor: } f_a = |f_r \bmod f_s|, \text{ folded to } [0, f_s/2]$$

THE Z-PLANE



zeros cancel the road · poles at 0.85 keep it narrow
dashed: the aliased reading of a 1.2 kHz sensor

ARCHITECTURE

The browser plays, the server judges

Browser

Vercel · React 19 · TanStack Start

- Every station's DSP, instantly
- Graphs and Web Audio
- Run state in localStorage



settings only

API

Vercel Python · FastAPI · Singapore

- Seeded dirty ingredients
- Rebuilds and judges the dish
- Points, tiers, leaderboard



SQL via the
pooler

Database

Supabase · PostgreSQL

- Chefs and sessions
- Every served attempt
- Scoring seasons

KEY DECISION

`browser → server: settings (bowl, cutoffs, dials, appliance, notch, poles)`

`server → browser: scores, stars, stages, ranks (never trusted from the client)`

Also in the game: the Signal Playground (10 labs), seasonal leaderboards, pause, countdown and sound.

IMPLEMENTATION

How a dish becomes a score

DISH MATCH (SERVER)

```
SNR = 10 · log10(Σt2 / Σ(t - p)2)  
corr = max over ℓ of (t * p)[ℓ] / (||t|| · ||p||)  
spec = cos∠(T_dB, P_dB), 40 dB window  
m = 40% · S + 35% · C + 25% · N  
S = (spec - 0.6) / 0.38, N = (SNR + 5) / 30, each clipped to 0–1
```

OVERALL SCORE

```
total = round(10 · score · d)  
        + round(min(2 · s_left, 300) · d)  
d = 0.8 · 1.0 · 1.25 · 1.5  
Easy → MasterChef; the leaderboard ranks this number.
```

FINAL SCORE

```
raw = m · (0.72 + 0.28 · prep / 100)  
score = 100 · (raw / 100)(1 / tol)  
stars at 92 / 80 / 66 / 50 / 32  
Washing scales the dish, so it cannot carry a ruined one.
```

LIVE MATCH (IN EVERY LAB)

```
r = Σ(x - μx)(y - μy) / √(Σ(x - μx)2 · Σ(y - μy)2)  
NRMSE = 1 - RMSE / (max y - min y)  
live = 50 · max(0, r) + 50 · NRMSE
```

RESULTS

Precision pays, mistakes cost

100%

perfect cooking matches the
reference, all 5 recipes

99–100%

the served dish after a
perfect oven and cart

93–98

server score for a perfect
finish

17–36

server score with the notch
left off

TESTED AND LIVE

187 frontend tests · **108** backend tests · **0** type or lint errors

wavekitchen.vercel.app

Thank you

Questions are welcome.

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